Physical Dynamics, Locked

From RG's Native Principles to the Einstein Branch: Selection, Exact Embedding, Attraction and Empirical Selection

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Abstract

This paper closes the physical-dynamics question as far as it can be closed. Every remaining premise is named, the mathematics carries every step, and the data select what the mathematics leaves open.

The principles. RG's own principles are lifted from the scalar network to the tensor:

- *relabeling-null:* descent makes coordinate relabelings invisible, i.e. gauge;
- *reciprocity:* the response is symmetric, i.e. variational;
- *balance:* the response operator is compatible with every conserved source.

The theorems.

- **Theorem A (linear selection).** Take the general local, Lorentz-covariant, parity-even, two-derivative operator on a symmetric tensor, with five coefficients. Any *two* of the three principles force it to be $cG^{(1)}_{\mu\nu}$; each alone leaves a two-dimensional family. Relabeling-nullity forbids a mass term. The linearized Ricci tensor is the canary: it is gauge invariant but not divergence-free.
- **Theorem B (Einstein branch).** In four dimensions, under a declared metric action class, $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$. Gauss–Bonnet is verified topological in four dimensions and dynamical in five; R^2 gives fourth-order equations.
- **Theorem C (exact embedding).** Every vacuum Einstein metric solves the broader $R + \alpha R^2 - 2\Lambda$ dynamics for every α. This is verified on Schwarzschild–de Sitter, whose Weyl curvature is nonzero.
- **Theorem D (global homogeneous attraction).** *Every* expanding spatially flat homogeneous vacuum solution converges to the de Sitter Einstein state, including solutions starting above the plateau. This strengthens the earlier sufficient condition $\mathcal{E}(0) < 1/(8\alpha)$ to the whole expanding class. 48 trajectories reach the state within 6×10^{-12}.
- **Theorem E (linear inhomogeneous attraction).** About the de Sitter Einstein state the scalaron decouples exactly and every Fourier mode decays at the predicted rate (agreement within 0.14%). Graviton modes freeze and are conserved. The Einstein sector is *normally attracting*: the extra degree of freedom dies, while GR's own degrees of freedom evolve conservatively, as Liouville requires.
- **Theorem F (weak field and selection).** $R + \alpha R^2$ adds a Yukawa force of strength $\frac{1}{3}$ and range $\sqrt{6\alpha}$, with $\gamma(r) = (3 - e^{-mr})/(3 + e^{-mr})$. Cassini bounds the range below 1.1×10^8 m; torsion balances exclude gravitational-strength Yukawa forces beyond $38.6\,\mu$m. The data select the Einstein branch at every tested scale.

What remains declared.

- the field content (one symmetric tensor);
- the Lorentzian cone;
- locality;
- the derivative order;
- $D = 4$;
- universal coupling;
- the anchors G and Λ.

Each is listed with its empirical support. None is hidden.

Evidence. Eight witness groups, symbolic or seeded, rerun byte-identically.

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1 What "closing physical dynamics" can mean

No-Go 1.1 (Liouville; Source of Gravity, No-Go 1.1). Hamiltonian dynamics has no attractor. GR's evolution is Hamiltonian, so no state of GR can attract other GR states.

Consequently, a closure cannot be "all geometries flow to one geometry". It can be four things, and this paper delivers each:

(1) a *selection* theorem: native principles force the field equation (Theorems A, B);

(2) an *embedding* theorem: GR is an exact sector of a broader dynamics (Theorem C);

(3) a *normal-attraction* theorem: the broader dynamics flows *onto* the GR sector, while GR's own degrees of freedom remain conservative (Theorems D, E);

(4) an *empirical selection*: data push the broader dynamics onto GR at every tested scale (Theorem F).

Obligation	Result here	Status
Operational state reconstruction	predictive quotient; metric from clocks and radar (Section 2)	PROVED, given anchors
Source characterization	balance on networks; conserved T (Section 3)	FORCED; tensor form declared
Dynamical admissibility	R1–R3 lifted to the tensor (Section 3)	RG-NATIVE
Source–response relation	Theorems A, B	PROVED (linear); CONDITIONAL (nonlinear)
Consistency and evolution	Bianchi; Theorems C–E	PROVED
Uniqueness and prediction	Theorem F; anchors; predictions from RG Gravitation	EMPIRICALLY SELECTED

2 Operational state and metric realization

The predictive quotient is the coarsest exact state (*From the Wall-Tap*, Proposition 3.5). For a static spherical region with calibrated areal radius, clock ratios $z = N/N_0$ and the proper radar time $T(r)$ measured on the emitter clock give:

Proposition 2.1 (Metric from records; P1)**.** *$T(r) = \frac{2}{c}\int_{r_0}^{r} B/z\, ds$, hence $B = \frac{c}{2}\, z\, T'$, and the metric is*

$$ds^2 = -z^2 c^2 dt_0^2 + \left(\tfrac{c}{2} z T'\right)^2 dr^2 + r^2 d\Omega^2.$$

A single finite radar interval leaves an exact nullspace: a perturbation of B/N proportional to $(r - r_0)(R - r)(r - \frac{r_0+R}{2})$ integrates to zero while changing the local geometry.

This is where geometry enters: as a *reconstruction* from records, not as a background.

3 RG's native principles, lifted to the tensor

Principle R1 (Relabeling-null)**.** By descent, quantities must not depend on representatives (GR Without Spacetime). Infinitesimal relabelings $x \mapsto x + \xi$ act on a perturbation of a flat background by $h \mapsto h + \partial\xi + \partial\xi$. An admissible response operator must annihilate such perturbations.

Principle R2 (Reciprocity)**.** The influence of A on B equals that of B on A. For a network this is a symmetric response matrix. For a continuum operator it is formal self-adjointness with respect to the pairing $\langle k, h\rangle = \int k^{\mu\nu} h_{\mu\nu}$, i.e. the response derives from a quadratic action.

Principle R3 (Balance)**.** Exchange conserves the total (Source of Gravity, Proposition 2.1). The response operator must accept every conserved source, i.e. be identically divergence-free: $\partial^\mu \mathcal{E}_{\mu\nu}[h] \equiv 0$.

Correction (P8a). For the scalar network, R1 and R2 do not imply stability: $L = -B^\top B$ is reciprocal and shift-blind, yet $\dot\phi = -L\phi$ is unstable. *Passivity*, $L = B^\top W B$ with $W \geq 0$, is a separate declaration. The Source of Gravity paper omitted it, and it is added here.

Declaration D1 (Field content and form)**.** One symmetric tensor $h_{\mu\nu}$ (the realized metric perturbation), a declared Lorentz cone η, locality, parity evenness, and response operators with exactly two derivatives. Zero-derivative terms are admitted only to test mass.

Theorem A (Linear selection; P2)**.** *Under D1, the general operator is*

$$\mathcal{E}_{\mu\nu} = A\,\Box h_{\mu\nu} + B(\partial_\mu v_\nu + \partial_\nu v_\mu) + C\,\partial_\mu\partial_\nu h + \eta_{\mu\nu}\big(D\,\partial\cdot v + E\,\Box h\big),$$

with $v_\nu = \partial^\mu h_{\mu\nu}$ *and* $h = \eta^{\mu\nu}h_{\mu\nu}$*.*

(a) R1 alone gives $B = -A$*,* $C = A$*,* $D = -E$*: a two-dimensional family, spanned by linearized* $R_{\mu\nu}$ *and* $\eta_{\mu\nu}R$*.*

(b) R3 alone gives $B = -A$*,* $C = -E$*,* $D = A$*: a two-dimensional family.*

(c) Any two of R1, R2, R3 give $B = -A$*,* $C = A$*,* $D = A$*,* $E = -A$*, i.e.*

$$\mathcal{E}_{\mu\nu} = -2A\,G^{(1)}_{\mu\nu}.$$

(d) R1 forbids mass terms $a_0 h_{\mu\nu} + b_0\eta_{\mu\nu}h$*: the mediator is massless.*

(e) Linearized $R_{\mu\nu}$ *satisfies R1 but not R3: gauge invariance alone does not give a conserved coupling (correction P8b to the Source of Gravity paper).*

Proof. For R1, apply $\mathcal{E}$ to $h = \partial\xi + \partial\xi$ with four arbitrary functions $\xi_\mu(x^0, \ldots, x^3)$ and require every coefficient of every independent third derivative to vanish. For R3, apply ∂^μ to $\mathcal{E}[h]$ for ten arbitrary functions $h_{\mu\nu}$ and do the same. For R2, compare $\langle k, \mathcal{E}h\rangle$ with $\langle \mathcal{E}k, h\rangle$ on the Fourier symbol over all pairs of basis tensors. Each yields a linear system in (B, C, D, E), and its solutions are as stated. The selected operator agrees component by component with $-2A$ times the linearized Einstein tensor built from the linearized Riemann tensor. Everything is computed symbolically (P2), with no hand-entered equations. This repairs the uploaded package's S06, whose equation list was written by hand. □

Reading. The scalar network had two native properties, shift-blindness and reciprocity, and they forced the Laplacian. The tensor has the same two properties, relabeling-blindness and reciprocity, and they force the linearized Einstein operator. Either one of them, together with balance, does the same. The masslessness of the mediator is a consequence, not an input.

4 The nonlinear Einstein branch

Declaration D2 (Action class)**.** The full dynamics derives from a local, relabeling-invariant action of one metric, with Levi-Civita connection, field equations of order at most two, regular at zero curvature, in $D = 4$. Matter enters through a separate action defining $T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\delta S_m/\delta g^{\mu\nu}$, and couples universally (D3).

Declaration D3 (Universal coupling)**.** All matter couples to the one metric.

Imported Theorem 4.1 (Lovelock; Deser [1, 2])**.** *In* $D = 4$ *the only symmetric divergence-free tensor built from* g *and its first two derivatives is* $aG_{\mu\nu} + bg_{\mu\nu}$*. Self-consistent coupling of the massless field of Theorem A to its own conserved stress–energy reconstructs the same equation.*

Theorem B (Einstein branch)**.** *Under D1–D3,*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \qquad \nabla^\mu G_{\mu\nu} \equiv 0 \Rightarrow \nabla^\mu T_{\mu\nu} = 0.$$

Checks (P3, FLRW minisuperspace with lapse).

- $\sqrt{-g}\,R$ gives second-order Euler–Lagrange equations.
- $\sqrt{-g}\,(R_{abcd}R^{abcd} - 4R_{ab}R^{ab} + R^2)$ gives identically vanishing equations in $D = 4$ (topological), but nonvanishing second-order equations in $D = 5$.
- $\sqrt{-g}\,R^2$ gives fourth-order equations.

This is exactly the Lovelock structure that makes $D = 4$ select Einstein–Hilbert. The contracted Bianchi identity was verified symbolically in the Source of Gravity paper (G8).

5 The broader dynamics and its exact Einstein sector

Relax D2 to allow fourth-order equations, with the simplest extension

$$S = \frac{1}{2\kappa}\int \sqrt{-g}\,(R - 2\Lambda + \alpha R^2)\,d^4x + S_m, \qquad \alpha > 0.$$

Its field equations and trace are

$$(1+2\alpha R)R_{\mu\nu} - \tfrac{1}{2}(R - 2\Lambda + \alpha R^2)g_{\mu\nu} + 2\alpha(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)R = \kappa T_{\mu\nu}, \qquad 6\alpha\Box R - R + 4\Lambda = \kappa T.$$

Theorem C (Exact embedding; P4)**.** *Every vacuum Einstein metric $R_{\mu\nu} = \Lambda g_{\mu\nu}$ solves the broader vacuum equations for every α.*

Proof. $R = 4\Lambda$ is constant, so the derivative terms vanish. Then $(1+8\alpha\Lambda)\Lambda g - \frac{1}{2}(2\Lambda + 16\alpha\Lambda^2)g = 0$.

Witness: for Schwarzschild–de Sitter, the full tensor equation vanishes symbolically for all α, the Kretschmann scalar exceeds its de Sitter value by $48M^2/r^6$ (nonzero Weyl curvature), and the same metric with $\Lambda \to 2\Lambda$ fails. □

6 Attraction onto the Einstein sector

In the Einstein frame, with $\kappa = 1$, the vacuum theory is GR plus a scalar $\chi = \sqrt{3/2}\ln\phi$, $\phi = 1 + 2\alpha R$. Its potential is

$$W(\chi) = \Lambda e^{-2b\chi} + \frac{(1 - e^{-b\chi})^2}{8\alpha}, \qquad b = \sqrt{2/3}.$$

With $z = e^{-b\chi}$, $W'(\chi) = bz\left[\frac{1-z}{4\alpha} - 2\Lambda z\right]$. Hence $W' < 0$ for $\chi < \chi_*$ and $W' > 0$ for $\chi > \chi_*$, where $\phi_* = e^{b\chi_*} = 1 + 8\alpha\Lambda$. Moreover $W_* = \Lambda/\phi_*$ is the global minimum, $W \to \infty$ as $\chi \to -\infty$, and $W \uparrow 1/(8\alpha)$ as $\chi \to \infty$.

Theorem D (Global homogeneous attraction; P5)**.** *Let $\Lambda > 0$. Every spatially flat homogeneous expanding vacuum solution, i.e.*

$$\ddot{\chi} + 3H\dot{\chi} + W'(\chi) = 0, \qquad H = +\sqrt{(\tfrac{1}{2}\dot{\chi}^2 + W)/3},$$

converges to $(\chi_*, 0)$ for any *finite initial energy. In Jordan variables, $\phi \to \phi_*$ and $H_J \to \sqrt{\Lambda/3}$: the de Sitter Einstein state.*

Proof. Let $\mathcal{E} = \frac{1}{2}\dot{\chi}^2 + W$.

1. $\dot{\mathcal{E}} = -3H\dot{\chi}^2 \le 0$, and $\mathcal{E} \ge W \ge W_* > 0$, so $H \ge H_{\min} = \sqrt{W_*/3} > 0$ for all time.
2. *Lower bound.* $W(\chi(t)) \le \mathcal{E}(0)$ and $W \to \infty$ as $\chi \to -\infty$, so χ is bounded below.
3. *Upper bound.* While $\chi > \chi_*$ we have $W' > 0$, so $\frac{d}{dt}\left(\dot{\chi}\, e^{\int 3H}\right) = -W' e^{\int 3H} < 0$. Hence, from any time t_0 at which χ enters or starts in that region, $\dot{\chi}(t) \le \max(\dot{\chi}(t_0), 0)\, e^{-3H_{\min}(t-t_0)}$, and $\chi(t) \le \chi(t_0) + \max(\dot{\chi}(t_0), 0)/(3H_{\min})$. With $|\dot{\chi}| \le \sqrt{2\mathcal{E}(0)}$ this gives
$$\chi(t) \le \max(\chi(0), \chi_*) + \sqrt{2\mathcal{E}(0)}/(3H_{\min}).$$
4. The orbit is therefore precompact. By LaSalle's invariance principle it approaches the largest invariant subset of $\{\dot{\mathcal{E}} = 0\} = \{\dot{\chi} = 0\}$. Invariance forces $W'(\chi) = 0$, i.e. $\chi = \chi_*$.
5. In Jordan variables, $H_J = \sqrt{\phi}\,(H - \frac{b}{2}\dot{\chi}) \to \sqrt{\phi_*}\sqrt{\Lambda/(3\phi_*)} = \sqrt{\Lambda/3}$.

Witness: 48 trajectories for $\alpha = 0.2, 1, 5$ and $\Lambda = 0.03$, 24 of them starting above the plateau (up to 6.2×). All converge to within 5.7×10^{-12}, with energy non-increasing and χ below the a priori bound. □

Note. Above the plateau the dynamics is slow roll, i.e. Starobinsky inflation. The uploaded package's condition $\mathcal{E}(0) < 1/(8\alpha)$ is sufficient but not necessary; Theorem D removes it.

Theorem E (Linear inhomogeneous normal attraction; P6)**.** *Linearize the full Einstein-frame theory about the de Sitter Einstein state (g = de Sitter with $H^2 = W_*/3$, $\chi \equiv \chi_*$).*

(a) Decoupling. *Because $\partial\chi_* = 0$ and $W'(\chi_*) = 0$, the scalar stress tensor contributes only $-W_*\delta g_{\mu\nu}$ at first order. Metric perturbations obey linearized GR with $\Lambda_{\text{eff}} = W_*$. The scalar obeys $\Box\delta\chi - m^2\delta\chi = 0$ with*

$$m^2 = W''(\chi_*) = \frac{1}{6\alpha\phi_*}, \qquad \frac{m^2}{H^2} = \frac{1}{2\alpha\Lambda} > 0.$$

(b) Scalar decay. *Each Fourier mode obeys $q'' + 3Hq' + (k^2e^{-2Ht} + m^2)q = 0$. Its energy $\frac{1}{2}(q'^2 + (k^2e^{-2Ht} + m^2)q^2)$ satisfies $\dot{} = -3Hq'^2 - Hk^2e^{-2Ht}q^2 \le 0$, and its late-time decay rate is*

$$\gamma = \tfrac{3}{2}H - \sqrt{\max(\tfrac{9}{4}H^2 - m^2, 0)} > 0.$$

(c) Gravitons are conserved. *Each polarization obeys $h'' + 3Hh' + k^2e^{-2Ht}h = 0$, which freezes at a nonzero constant. There is no attraction within the GR sector.*

(d) Jordan frame. *$\delta g^J = e^{-b\chi_*}(\delta g^E - b\, g^E\delta\chi)$, so the Jordan perturbation converges to a GR perturbation.*

Witness. Nine scalaron modes ($\alpha = 1, 5, 20$; $k = 0, 0.5, 3$) give fitted rates equal to the predicted rates to within $2.1 \times 10^{-3}H$, i.e. $1.5H$ in the underdamped case and $0.3098H$ in the overdamped case, with energy monotone. The underdamped fit uses the exactly exponential Lyapunov function $\frac{1}{2}((q' + \frac{3}{2}Hq)^2 + \omega^2q^2)$. Nine graviton modes freeze to within relative change 10^{-6}.

Corollary 6.1 (What attracts what)**.** *In the space of dynamics containing GR, the Einstein sector is a normally attracting invariant set at linear order (and globally for homogeneous expanding data): the non-Einstein degree of freedom decays, while GR's degrees of freedom are conserved. This is the only kind of "GR attractor" compatible with Liouville, and it holds.*

Remark 6.2 (The limit $\alpha \to 0$ is not uniform). As $\alpha \to 0$, $m \to \infty$: the scalaron oscillates ever faster. GR is recovered for well-prepared data, or after the decay of Theorem E. It is not recovered uniformly for arbitrary initial data.

7 Weak field, prediction and empirical selection

Theorem F (Weak field of the broader dynamics; P7)**.** *For a static point mass in $R + \alpha R^2$, with $\Lambda = 0$ and isotropic gauge $ds^2 = -(1 + 2\Psi)dt^2 + (1 - 2\Phi)d\mathbf{x}^2$,*

$$\Psi = -\frac{GM}{r}\Big(1 + \frac{e^{-mr}}{3}\Big), \qquad \Phi = -\frac{GM}{r}\Big(1 - \frac{e^{-mr}}{3}\Big), \qquad m = \frac{1}{\sqrt{6\alpha}}.$$

This solves the linearized vacuum equations exactly. Its curvature is $R^{(1)} = 2GMm^2e^{-mr}/r$, the Yukawa solution of $(1 - 6\alpha\nabla^2)R = \kappa\rho$ for a point source with $\kappa = 8\pi G$, so the solution is normalized to the source. Consequently:

- *the Newtonian potential carries a Yukawa term of strength $\frac{1}{3}$ and range $\lambda = \sqrt{6\alpha}$;*

- $\gamma(r) = \Phi/\Psi = (3-e^{-mr})/(3+e^{-mr})$, *which is* $\frac{1}{2}$ *at short range and* 1 *(GR) at long range;*
- *the GR solution* $\Psi = \Phi = -GM/r$ *has* $R^{(1)} = 0$*, and so cannot satisfy the trace equation with a point source unless* $\alpha = 0$*.*

Empirical selection.

- **Cassini.** $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [6]. Since $\gamma - 1 = -2e^{-mb}/(3 + e^{-mb}) < 0$, the 2σ lower end -2.5×10^{-5} at impact parameter $b \approx 1.6 R_\odot$ requires $mb \geq 10.19$, i.e. a range below 1.09×10^8 m and $\alpha \leq 2 \times 10^{15}\,\mathrm{m}^2$. This is a weak bound.
- **Torsion balance.** Gravitational-strength Yukawa forces ($|\alpha_Y| = 1$) are excluded beyond $38.6\,\mu$m at 95% confidence [7]. The R^2 Yukawa has $\alpha_Y = \frac{1}{3}$, whose exclusion range is of the same sub-millimetre order. That range is not read off here. If it is taken as $38.6\,\mu$m, then $\alpha \lesssim 2.5 \times 10^{-10}\,\mathrm{m}^2$.
- **Mediator.** Masslessness and spin 2 are supported by:
  - the tensor speed, $|c_T/c - 1| \leq 3.1 \times 10^{-15}$ (GW170817);
  - quadrupole radiation, with ratios 0.9983±0.0017 and 0.999963±0.000063 (B1913+16, J0737);
  - $\gamma = 1$.

  All are from GR Conditional Naturality. These certify D1 and D3 empirically. They do not derive them.

Reading. The broader dynamics differs from GR only below the scalaron range, and data push that range below $\sim 10^{-4}$ m. At every tested scale the Einstein branch is selected. With κ anchored once (Kepler), the predictions follow: Mercury $42.98''$/cy, deflection $1.751''$, the GPS clock offset and the Pound–Rebka shift (RG Gravitation).

8 Corrections log

	Before	After
P8a	Source of Gravity: shift-blind + reciprocal ⇒ attracting Laplacian	passivity $W \geq 0$ must be declared (counterexample $L = -B^\top B$)
P8b	Source of Gravity G10: gauge invariance ⇒ conserved source	needs a second property, reciprocity or identical divergence-freeness (Theorem Ae)
S06	uploaded package: hand-written coefficient equations	derived symbolically from generic functions (Theorem A)
D01	uploaded package: attraction for $\mathcal{E}(0) < 1/(8\alpha)$	every expanding homogeneous solution (Theorem D)
—	"$\alpha \to 0$ recovers GR"	only for well-prepared or decayed data (Remark 6.2)

9 Final ledger

Item	Status	Basis
Metric from clock ratios and radar	PROVED	P1
Relabeling-null, reciprocity, balance	RG-NATIVE	descent; network symmetry; exchange
Passivity (scalar network)	DECLARED	P8a
Linear operator $= cG^{(1)}$; massless mediator	PROVED	Theorem A
Symmetric tensor field, Lorentz cone, locality, two derivatives	DECLARED; empirically supported	c_T, quadrupole, γ
$D = 4$; universal coupling	DECLARED; empirically supported	Eötvös-type and γ tests
Nonlinear Einstein equation	IMPORTED (Lovelock, Deser) + checks	Theorem B, P3
GR as exact sector of $R + \alpha R^2$	PROVED	Theorem C
Global homogeneous attraction	PROVED	Theorem D
Linear inhomogeneous normal attraction	PROVED	Theorem E
Weak field, $\gamma(r)$	PROVED	Theorem F
Data select the Einstein branch	EMPIRICAL	Cassini; torsion balance
G, Λ	ANCHORS	one measurement each
Field content, cone and $D = 4$ from network microdynamics	OPEN	—
Nonlinear inhomogeneous stability; attraction with matter	OPEN	—
Quantum renormalization flow	NOT ADDRESSED	—

The locked statement. *RG's native principles (relabeling-nullity, reciprocity, balance), applied to one symmetric tensor with a Lorentz cone and two derivatives, force the linearized Einstein operator and a massless mediator. In four dimensions the nonlinear completion is Einstein's equation. Embedded in the broader $R + \alpha R^2$ dynamics, the Einstein sector is exact, attracts every expanding homogeneous solution, and attracts the extra degree of freedom linearly in general, while GR's own degrees of freedom remain conservative. The data restrict the extra sector to sub-millimetre range, selecting the Einstein branch at every tested scale.*

What this closes. The physical-dynamics gap of the RG print, i.e. *which law and why*, is closed up to the listed declarations. Each declaration is empirically certified, not derived. Deriving the declarations themselves from a microscopic network remains open.

10 Reproduction and limits

- `rg_physical_dynamics.py`, with `curv.py`, runs P1–P8. P1–P4, P6(a), P7 and P8 are symbolic; P5 and P6(b,c) use deterministic integration. Reruns are byte-identical.
- *Limits.*
  - Theorem A assumes D1.
  - The P3 checks are on minisuperspace; the general Lovelock and Deser results are imported.

– Theorems D and E are vacuum results: homogeneous and globally nonlinear for D, linear and inhomogeneous for E.

– The torsion-balance number is for $|\alpha_Y| = 1$.

– No new data were fitted.

References

[1] D. Lovelock, The Einstein tensor and its generalizations, *J. Math. Phys.* 12 (1971) 498.

[2] S. Deser, Self-interaction and gauge invariance, *Gen. Rel. Grav.* 1 (1970) 9.

[3] A. A. Starobinsky, A new type of isotropic cosmological models without singularity, *Phys. Lett. B* 91 (1980) 99.

[4] K. S. Stelle, Classical gravity with higher derivatives, *Gen. Rel. Grav.* 9 (1978) 353.

[5] J. P. LaSalle, Some extensions of Liapunov's second method, *IRE Trans. Circuit Theory* 7 (1960) 520.

[6] B. Bertotti, L. Iess, P. Tortora, A test of general relativity using radio links with the Cassini spacecraft, *Nature* 425 (2003) 374.

[7] J. G. Lee, E. G. Adelberger, T. S. Cook, S. M. Fleischer, B. R. Heckel, New test of the gravitational $1/r^2$ law at separations down to $52\,\mu$m, *Phys. Rev. Lett.* 124 (2020) 101101.

[8] M. Rada, *Resolution Geometry* (print edition) and companions: Source of Gravity; From the Wall-Tap; GR Without Spacetime; GR Conditional Naturality; RG Gravitation; Monotonicity and the Lorentz Sign, 2026.